than their actuarially fair values. Milevsky and Posner (2001) value the insurance guarantees in annuities and find these are worth between 0.01% and 0.10%, but the insurance industry charges 1.15%.[56] Mitchell et al. (1999) estimate the value of annuities offered by life insurance companies and find that it's 80 to 90 cents for each dollar. However, Mitchell et al. show that the difference between fair value and purchase price has shrunk over time.

Even if annuities are fairly priced for an average policyholder, they can be expensive for certain segments of the population. Brown, Mitchell, and Poterba (2002) report a male buying an annuity at age sixty-five would get 91 cents of value for every dollar put in, but a more general member of the public would be only getting 81 cents of value. This is because insurance companies selling annuities face . . .

Adverse Selection

Early in Jane Austen's *Sense and Sensibility*, Mrs. Dashwood convinces her husband not to help the family of our heroines (Elinor, Marianne, and Margaret), explaining: "People always live forever when there is an annuity to be paid them." Insurance companies take into account adverse selection when pricing annuities to avoid bankrupting themselves by paying people who "always live forever."

Perhaps there is a spiraling negative interaction of adverse selection with demand: adverse selection drives up the price of annuities, this leads to lower demand, which in turn leads to greater adverse selection, and so on. Chalmers and Reuter (2012b) rule out this explanation in a study of the Oregon Public Employees Retirement System. They do not find that participants in Oregon's pension plan buy more annuities when annuity prices fall, which must point to other investor-specific characteristics driving (lack of) annuity demand. Getting value for money is also not an issue in Oregon's system, as the lucky participants receive more than $1 in benefits in annuities for each $1 that they buy.

Framing

Consumers often view annuities as gambles. Benartzi, Previtero, and Thaler (2011) argue that consumers view annuities in an "investment" mindset, or *frame*, rather than in a "consumption" mindset. They come to an annuity decision with the question, "Will I live long enough for this investment to pay off?" The risk is that all the (usually considerable) money placed in the annuity, earned with blood, sweat, and tears, vanishes with early death. The investor benefits from guaranteeing his consumption each year (subject to credit risk) but does not view the annuity in terms of consumption benefits. Changing the frame can enhance demand.

[56] The basic benefit is that at least the original investment will be returned upon death. This is a put option with a strike price equal to the purchase price of the annuity. There are many variants, see They're Complicated.